

Customer Shopping Behaviour Analysis

Python | Pandas | SQL Server | SSMS | Power BI | Jupyter Notebook

1. Project Overview

This project analyses a retail dataset of 3,900 purchase records across multiple product categories to understand how customers shop, what they buy, how discounts relate to purchasing, and how subscription status and previous purchases describe different customer groups. The workflow combines data preparation, SQL-based business analysis and interactive visualisation to support data-driven retail decisions.

- Explore and prepare the dataset using Python and Pandas.
- Store and analyse the prepared data in SQL Server using SSMS.
- Answer ten business questions with SQL queries.
- Present shopping patterns through a Power BI dashboard and derives potential recommendations.

2. Business Problem

A retail company wants to improve sales, customer satisfaction and long-term loyalty by understanding its customers' shopping habits: popular products, discount usage and ratings, spending patterns across customer groups, and the link between purchase history and subscription status.

Main question: How can the company use customer shopping data to identify trends, improve engagement, and optimise marketing and product strategies?

3. Dataset

The dataset is structured tabular retail data with 3,900 rows and 18 columns. The only missing values are 37 entries in review_rating.

Field group	Columns and purpose
Customer	customer_id, age, gender, location – customer-level and demographic analysis
Product	item_purchased, category, size, color, season – product, preference and seasonal analysis
Transaction	purchase_amount_usd, payment_method, shipping_type – purchase amount and payment/shipping comparisons
Promotions	discount_applied, promo_code_used – discount and promo-code usage
Behaviour	previous_purchases, frequency_of_purchases, subscription_status, review_rating – segmentation, repeat purchases, ratings

4. Tools and Technologies

Tool	Purpose
Python, Pandas, Jupyter Notebook	Data exploration, cleaning, preparation and analysis
Microsoft SQL Server, SSMS	Database storage, query execution and result inspection
SQL	Answering the ten business questions
Power BI	Interactive dashboards and visualisation
GitHub	Version control and project documentation

5. Python Data Cleaning and Preparation

- **Loading and exploration:** loaded the data into a DataFrame and reviewed structure with df.info() and df.describe().
- **Missing values:** found 37 missing review_rating values and filled each with the median rating of its product category.
- **Column renaming:** converted column names to snake_case for readability in Python and SQL.
- **Feature engineering:** created age_group for comparative analysis and purchase_frequency_days from the purchase-frequency field (interpretation depends on the mapping used).
- **Discount review:** checked whether discount_applied and promo_code_used carry overlapping information.
- **Database integration:** connected Python to SQL Server and loaded the prepared dataset for SQL analysis.

6. SQL Business Analysis

Ten business questions were answered in SQL Server, covering spending, ratings, discount usage, subscriptions, segmentation and purchase history.

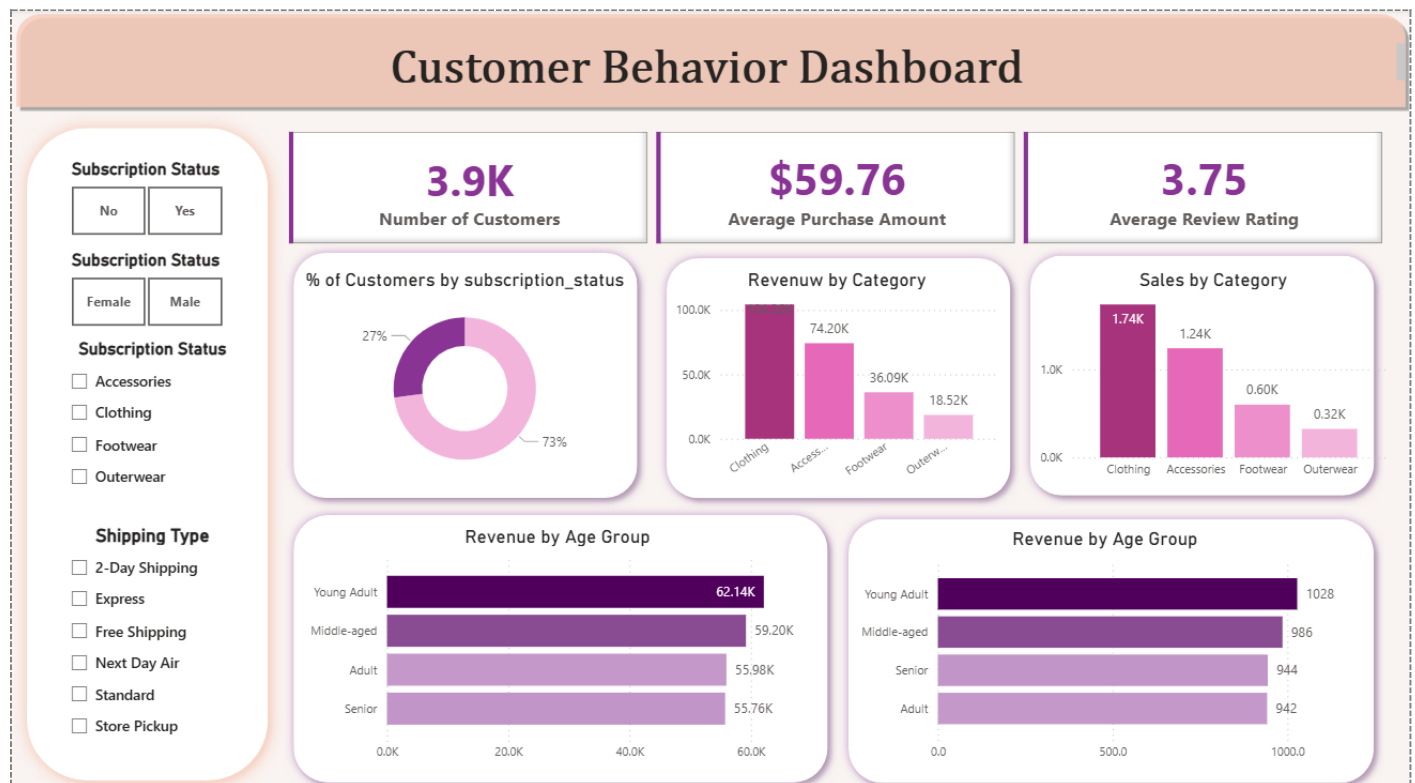
No.	Business question	Objective
1	Revenue by gender	Compare total recorded purchase amount for male and female customers
2	High-spending discount users	Find discounted purchases at or above the overall average amount
3	Top five products by rating	Identify the five products with the highest average review rating
4	Shipping type comparison	Compare average purchase amount for Standard vs Express shipping
5	Subscribers vs non-subscribers	Compare average and total spending by subscription status
6	High discount-usage products	Find the five products with the highest share of discounted purchases
7	Customer segmentation	Group customers as New, Returning or Loyal by previous purchases
8	Top three products per category	Find the three most frequently purchased products in each category
9	Repeat buyers and subscriptions	Examine subscription status where previous purchases exceed five
10	Revenue by age group	Calculate total purchase amount across age groups

SQL concepts used: SELECT/WHERE, aggregates (SUM, AVG, COUNT), GROUP BY/ORDER BY, CASE logic, subqueries, and ranking and percentage calculations where applicable. Query results should be added to this report only after the executed outputs are verified.

7. Power BI Dashboard

The dashboard presents customer purchasing patterns interactively so they are easier to explore and compare. It covers:

- Purchases by product category and customer demographics
- Discount and promo-code usage, and payment-method preferences
- Customer review ratings, seasonal purchasing patterns
- Subscription status and previous-purchase behaviour



8. Business Recommendations

These are potential recommendations based on the types of analysis performed; their effectiveness must be evaluated against actual results and additional business data.

Area	Recommendation
Subscriptions	Explore exclusive benefits, targeted offers and incentives to encourage subscriptions
Customer loyalty	Consider loyalty programmes, rewards and personalised offers for repeat buyers
Discount policy	Evaluate discount usage alongside purchase amount; add profit-margin data where available
Product positioning	Feature highly rated, frequently purchased products in campaigns, subject to verified results
Targeted marketing	Use demographics and purchasing patterns to tailor campaigns by customer group
Seasonal planning	Use seasonal patterns to guide promotions, inventory and product availability

9. Repository Structure

Suggested structure (adjust to the files actually uploaded):

Folder / file	Purpose
data/	Dataset information and, if permitted, data files
notebooks/	Jupyter Notebook with the analysis
python/	Data preparation scripts (e.g. data_cleaning.py)
sql/	SQL queries for the ten business questions
powerbi/ and images/	Power BI .pbix file and dashboard screenshots
report/ and README.md	Project report and documentation

10. How to Run the Project

- **Clone:** git clone <your-github-repository-url>, then cd Customer-Shopping-Behaviour-Analysis.
- **Install:** pip install pandas sqlalchemy pyodbc jupyter.
- **Run Python:** open the notebook with jupyter notebook and run the cells in order, or run python/data_cleaning.py.
- **Configure SQL Server:** connect in SSMS and set your own server, database and authentication; use environment variables rather than hard-coded credentials.
- **Analyse in SQL:** load the prepared data, run the queries in SSMS and review results before documenting findings.
- **Open Power BI:** open the .pbix file in Power BI Desktop and refresh the data source if required.

11. Key Learnings and Future Improvements

Key learnings: data exploration and cleaning in Python; SQL filtering, aggregation, grouping and conditional logic; customer segmentation by previous purchases; Power BI visualisation; translating analytical questions into potential retail recommendations.

Future improvements: time-based trend analysis with transaction dates; sales-channel comparison (online vs offline); profitability analysis with cost and margin data; advanced segmentation such as clustering; automated reporting with scheduled refresh. These are potential enhancements, not part of the current analysis.

12. Conclusion

This project combines Python, Pandas, SQL Server and Power BI to explore customer shopping behaviour. Python prepares the data, SQL enables structured analysis, and Power BI presents the results visually, demonstrating a practical end-to-end analytics workflow for identifying opportunities in retail decision-making.